Hirak Mondal

Senior Engineer - Samsung Research,Bangalore
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Work Experience

Senior Engineer(Speech AI) - Samsung Research, Bangalore (*Aug'22-present*)

- Developed and optimized on-device ASR models deployed in Samsung S23–S26 flagship devices, improving accuracy and latency under strict compute constraints.
- Led end-to-end development of en-GB and en-AU ASR models—used globally across millions of devices.
- Working on SOTA research ideas for top-tier speech conferences like Interspeech and ICASSP.

Academic Qualifications

Indian Institute of Technology, Kanpur — *Kanpur, India*
M.Tech, Computer Science and Engineering — *Graduation Date (July, 2022)*
GPA: 8.28/10

University of Engineering and Management — *Kolkata, India*
B.Tech, Computer Science and Engineering — *Graduation Date (April, 2020)*
GPA: 9.17/10

Publications

- **Probing LoRA-to-LoRA Cross-Lingual Transfer for Unseen Low-Resource Conditions in Whisper-Based ASR**
  - Designed and evaluated a novel **LoRA-to-LoRA** transfer learning paradigm for multilingual Whisper ASR, preserving pretrained parameters while enabling efficient cross-lingual adaptation.
  - Engineered a language donor selection framework leveraging genealogical and orthographic similarity measures to maximize transfer effectiveness for under-resourced languages.
  - Achieved up to **17%** relative WER improvement on low-resource ASR benchmarks and validated the effectiveness of donor-informed initialization across multiple language pairs.
  - Work accepted at **Interspeech, 2026**.
- **Teacher-Free Knowledge Distillation for Short-Utterance Spoken LID**
  - Investigated failure modes of short-utterance Spoken Language Identification (LID) systems, revealing that **36.94%** of misclassifications on 2-second inputs stem from out-of-scope phenomena such as non-speech segments, named entities, filler words, and overlapped speech.
  - Developed a teacher-free knowledge distillation (TF-KD) framework with online label smoothing that leverages logits from correctly classified samples as soft targets, eliminating the need for a separately trained teacher network.
  - Proposed enhancements including **dynamic distillation weighting, conditional label updates, and entropy-based soft-label generation**, achieving consistent Cavg improvements across both same-corpora and cross-corpora short-duration LID evaluations.
  - Work accepted at **Interspeech, 2025**. PaperLink
- **Rethinking Organization Entity Modeling in End-to-End Acoustic Named Entity Recognition**
  - Investigated structural limitations of end-to-end acoustic NER models for organization entities and established the importance of explicit span-level supervision for robust entity recognition.
  - Proposed a novel SCEL (Structure-Constrained Entity Learning) objective alongside targeted semantic augmentation and boundary-aware training to improve organization entity detection in speech.
  - Advanced the state of speech-based named entity recognition by delivering substantial improvements in organization recognition accuracy without compromising transcription performance, surpassing several prior acoustic NER baselines.
  - Work accepted at **Interspeech, 2026**
- **Spoken Language Diarization using Source-Domain Aggregative Weakly-Supervised Language Labels**
  - Investigated the interaction between language identification and diarization in multilingual conversational speech, demonstrating the value of domain-specific language modeling for improving SLD performance.
  - Designed a novel **weakly supervised pseudo-labeling framework (ARPM)** to convert RTTM annotations into high-quality language labels for training Whisper-based language identification models without manual language annotations.
  - Advanced the state of spoken language diarization on the DISPLACE 2024 benchmark, achieving substantial DER improvements and performance comparable to top challenge systems while using significantly less training data and model complexity.
  - Work accepted at **SPCOM, 2026**

Technical Skills

- **Languages:** Python (Primary), C++, Java
- **ML Frameworks**: PyTorch, Hugging Face, Fairseq
- **Cloud/Tools**: AWS, Docker, Git
- **Speech AI**: ASR, LID, SLD, KD, Transformer models (Whisper, wav2vec2)
- **Specializations**: On-Device ASR, LoRA, Agentic AI, Speech Processing

Awards & Scholastic Achievements

- **Samsung Excellence Award** for commercialisation of enGB model on flagship S-series devices.
- **Samsung Project Incentive Award** for enGB ASR Project.
- Secured **AIR 111** in JEST 2020 Theoretical Computer Science Examination.